

# Deep inelastic scattering on polarized lithium: the physics case for ${}^6\text{Li}$ and ${}^7\text{Li}$ beams at the Electron-Ion Collider

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**Abstract.** Polarized  ${}^6\text{Li}$  and  ${}^7\text{Li}$  beams would give the Electron-Ion Collider (EIC) spin-polarized nuclei that are light enough for ab initio computation yet carry genuine nuclear structure:  ${}^6\text{Li}$  is a spin-1 nucleus well described as a deuteron bound to an inert  $\alpha$  particle,  ${}^7\text{Li}$  its spin-3/2 partner, an  $\alpha$  plus a triton. We review why lithium is the practical choice — an alkali atom that optical pumping polarizes, a pair of cluster nuclei whose spin sits on the light cluster, and ions whose spin dynamics in the ring are deuteron-like — and why the physics requires both polarization and a collider. Starting from the spin-1 formalism of inclusive deep inelastic scattering we develop the four measurements the programme is built around: the tensor structure function  $b_1$ , the double-helicity-flip structure function  $\Delta$  (nuclear gluonometry), the polarized EMC effect on the bound proton of  ${}^7\text{Li}$ , and spectator-tagged scattering on the embedded deuteron. For each we state what the beam must provide and what one EIC year ( $10\text{ fb}^{-1}$  per nucleon) reaches. Tensor polarization is the scarce resource, the transverse spin direction the natural one, and the spectator-tagged and coherent channels the ones that fix the far-forward requirements.

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## 1 · Introduction

The EIC project scope provides polarized electrons, protons and  ${}^3\text{He}$ ; polarized deuterons are a later addition, and no heavier polarized ion appears in the conceptual design or the Yellow Report [1,2]. The EPIOS white paper makes the case for polarized  ${}^6\text{Li}$  and  ${}^7\text{Li}$  as "a desirable addition", listed with  ${}^2\text{H}$  and  ${}^3\text{He}$  as critical technology [3], and the ANL source development specifies the beam: an atomic-beam source with optical pumping at 671 nm feeding EBIS charge breeding, with source targets  $P_z \geq 0.90$ ,  $P_{zz} \geq 0.80$  and  $\geq 10^{13}$  atoms/s [4]. Four measurements motivate the effort, each addressing a gap in the literature: no  $b_1$  prediction or measurement exists for any nucleus beyond the deuteron; no gluon-transversity projection exists for any EIC target; no study of  $\alpha$ -tagged deep inelastic scattering (DIS) on lithium exists; and no lithium-beam luminosity or polarization parameter set has been published.

This note is the physics case behind those measurements, written for colleagues joining the programme. Section 2 argues for lithium as an atom, as a nucleus and as a stored ion; Section 3 for polarization and for a collider; Section 4 develops the spin-1 formalism; Section 5 the four observables; Section 6 collects what each requires of the beam and what the programme's simulations project for one EIC year. The projections themselves — rates, bins, estimators and the reconstructed-level results — are the subject of the companion report [5].

## 2 · Why lithium

### 2.1 · The atom: optical pumping and a proven source

Lithium is the lightest alkali metal and its atom is hydrogen-like: a single 2s valence electron whose D lines near 671 nm (670.96 and 670.98 nm in vacuum for  ${}^7\text{Li}$ ) lie within the range of commercial single-frequency diode lasers, and whose ground-state hyperfine splittings — 228.2 MHz for  ${}^6\text{Li}$ , 803.5 MHz for  ${}^7\text{Li}$  [6] — are small enough for one laser with an electro-optic sideband to address both hyperfine levels. Circularly polarized light pumps the atoms into a single magnetic substate, the hyperfine coupling transfers the electron polarization to the nucleus, and radio-frequency transitions between Zeeman sublevels set any combination of  $P_z$  and  $P_{zz}$  [4]. Lithium's low ionization energy (5.39 eV [6]) eases ionization, and fully stripped  $\text{Li}^{3+}$  is within reach of the EBIS being commissioned to charge-breed the polarized  ${}^3\text{He}^{++}$  beam, whose design goal is  $2 \times 10^{11}$  ions per pulse at  $\geq 70\%$  polarization [7].

Both isotopes are stable and abundant (7.6%  ${}^6\text{Li}$ , 92.4%  ${}^7\text{Li}$ ), and they are the only stable isotopes of one element with  $J = 1$  and  $J = 3/2$ : the stable spin-1 nuclei are exactly  ${}^2\text{H}$ ,  ${}^6\text{Li}$  and  ${}^{14}\text{N}$ , and the stable spin-3/2 nuclei below  $A = 30$  are  ${}^7\text{Li}$ ,  ${}^9\text{Be}$ ,  ${}^{11}\text{B}$ ,  ${}^{21}\text{Ne}$  and  ${}^{23}\text{Na}$  [8]. One source, one laser system and one injector therefore serve a spin-1 and a spin-3/2 programme. EPIOS's list of where an exotic-glue discovery is most likely — "spin-1  ${}^6\text{Li}$  or  ${}^{14}\text{N}$ , or spin-3/2  ${}^7\text{Li}$  or  ${}^{23}\text{Na}$ " [3] — reduces, once optical pumping is required, to lithium and sodium —  ${}^{14}\text{N}$  is not an alkali and has no atomic-beam polarization route, and boron's electronic structure makes its optical pumping a development project of its own [3] — and lithium is the lighter, cluster-structured, ab initio-computable one.

The source physics has a forty-year history. The Heidelberg tandem delivered  ${}^6\text{Li}^{3+}$  at vector polarization 0.57–0.65 in 1977 [9]; by 1987 its atomic-beam source reached  $P_z, P_{zz} \geq 0.85$  and accelerated  ${}^7\text{Li}$  at  $P_{zz} = 0.69$  on target [10]; the optically pumped Florida State source delivered more than 150 nA of  ${}^6\text{Li}^{3+}$  at  $P_{yy} = -1.33 \pm 0.02$  with no polarization loss in the linac [11]; and polarized  ${}^6\text{Li}$ ,  ${}^7\text{Li}$  and  ${}^{23}\text{Na}$  beams served two decades of nuclear-reaction studies [12]. New for the EIC are scale — two to three orders of magnitude beyond these sources — and transport: preservation of the polarization through charge breeding and acceleration beyond 100 GeV/u [3,4].

## 2.2 · The nuclei: the spin sits on a light cluster

The two stable lithium isotopes are the lightest nuclei beyond  $A = 3$  in which a polarized light cluster is bound to an inert core: in  ${}^6\text{Li}$  an  $\alpha$  particle ( $J = 0, T = 0$ ) and a deuteron-like cluster bound by 1.47 MeV against  $\alpha+d$  breakup, in  ${}^7\text{Li}$  an  $\alpha$  and a triton bound by 2.47 MeV [8]. Variational Monte Carlo (VMC) with realistic two- and three-nucleon forces makes this quantitative: the  $\alpha$ -d overlap of the  ${}^6\text{Li}$  ground state integrates to  $N_{\alpha d} = 0.86$  (0.846 S-wave, 0.017 D-wave) and the  $\alpha$ -t overlap of  ${}^7\text{Li}$  to  $N_{\alpha t} = 1.00$  [13]; the no-core shell model with continuum gives an asymptotic D/S ratio of  $-0.027$  for the  $d+\alpha$  configuration [14]. In a fully polarized  ${}^6\text{Li}$  the  $\alpha$  contributes nothing and the two valence nucleons carry the spin: six-body VMC gives 1.924 spin-up and 1.076 spin-down protons (and neutrons) in the  $M = 1$  state, an effective polarization  $N_{\uparrow} - N_{\downarrow} = 0.85$  per cluster nucleon [13,15]; three-body  $\alpha+n+p$  models, in which  $\alpha+d$  accounts for 60–70% of the wave function, give above 90% [16]; and E155, which used  ${}^6\text{LiD}$  as its deuteron target, adopted 87% for the effective deuteron polarization [17].  ${}^6\text{Li}$  is therefore a polarized deuteron in the field of an  $\alpha$  particle, whose tensor observables can be set against the free deuteron's to isolate the medium effect. The only other stable spin-1 ground state,  ${}^{14}\text{N}$ , shares its spin among p-shell nucleons and lies beyond the reach of exact quantum Monte Carlo ( $A \leq 12$ ) [18].

In  ${}^7\text{Li}$  the spin is carried by the unpaired proton of the triton cluster, the neutrons being paired [19]. VMC gives proton numbers 1.934 up and 1.066 down and neutron numbers 1.981 and 2.019 [13], i.e.  $P_p = 0.87$  and  $P_n = -0.04$  (0.866 and  $-0.037$  as quoted in the JLab proposal [20]); the shell model already gives  $P_p = 13/15$ . Polarized  ${}^7\text{Li}$  is thus to the proton what polarized  ${}^3\text{He}$  is to the neutron, with the property, decisive for the polarized EMC effect, that the proton sits in a nucleus of normal density. Cloët, Bentz and Thomas set the selection criteria —  $|P_n| \ll |P_p|$  and light, because  $g_1^A$  is carried by the valence nucleons and is suppressed relative to  $F_2^A$  by roughly  $1/A$  — and list  ${}^7\text{Li}$  among the candidates "where the nuclear polarization is largely dominated by the valence proton" [21]; the gluon-spin study names  ${}^7\text{Li}$  "the most promising case for experimental investigation" [22].

Two further properties matter. The cluster structure makes the nuclei computable: Green's function Monte Carlo with AV18+IL7 reproduces both binding energies ( $-31.93$  vs  $-31.99$  MeV for  ${}^6\text{Li}$ ,  $-39.15$  vs  $-39.25$  MeV for  ${}^7\text{Li}$ ), the charge radii to within 3% and, with two-body currents, the magnetic moments ( $0.835$  vs  $0.822 \mu_N$ ;  $3.24$  vs  $3.256 \mu_N$ ) [18,19], so the nuclear corrections entering every extraction — effective polarizations, momentum distributions, spectral functions — carry controlled uncertainties rather than model spread. The quadrupole moment of  ${}^6\text{Li}$  is the exception: the measured  $Q = -0.0806 \text{ fm}^2$ , against  $+0.286 \text{ fm}^2$  for the free deuteron [8], arises from a delicate cancellation between the deuteron's own moment and the D-wave part of the  $\alpha$ -d relative motion [15,23], and theory reproduces only its smallness. That smallness is put to use in §5.2, where the near-sphericity of  ${}^6\text{Li}$  becomes a null test for the coherent channel;  ${}^7\text{Li}$ , whose p-shell triton orbit gives  $Q = -4.00 \text{ fm}^2$  [8], is the deformed counterpart. The  $\alpha+d$  versus  $\alpha+t$  comparison changes only the cluster.

### 2.3 · The ion: deuteron-like spin dynamics and deuteron kinematics

In the ring,  ${}^6\text{Li}$  is the easy case. Its gyromagnetic anomaly,  $G \approx -0.18$ , is as small as the deuteron's ( $-0.14$ ), so it crosses only 81 depolarizing resonances on the way to top energy, against 63 for the deuteron and 2457 for  ${}^3\text{He}$  [24]; like the deuteron it is handled with a partial solenoid snake and tune jumps, full Siberian snakes being impractically long for so small a  $G$  [3,24], so the deuteron techniques [25] carry over.  ${}^7\text{Li}$  ( $G \approx +1.52$ ) crosses 573 resonances and needs partial snakes with jump quadrupoles [3,24].

The energies follow from the collider, not from the ion. The hadron and electron rings must share a revolution period, so the hadron Lorentz factor is fixed by the circumference and the magnets supply whatever rigidity that  $\gamma$  demands, up to the 917 T·m cap: the ion menu is  $\gamma$ -matched to the proton's, an isolated point at  $\approx 41$  GeV/u and a continuous band from  $\approx 99$  GeV/u up to the species' rigidity limit of  $275 \text{ GeV} \times Z/A$  [3,26]. For  ${}^6\text{Li}$  that is 40.8 GeV/u and 99.5–137.5 GeV/u, for  ${}^7\text{Li}$  40.8 and 99.5–117.9 GeV/u. Because  ${}^6\text{Li}$  has  $A/Z = 2$  it runs at the same per-nucleon energy and magnet settings as the deuteron, so  $e+d$  and  $e+{}^6\text{Li}$  can be compared at identical kinematics. The stable spin direction in the arcs is vertical, so transverse polarization is native at the interaction point while longitudinal polarization needs spin rotators whose lithium settings are undefined [1,3].

## 3 · Why polarization, and why a collider

### 3.1 · Polarization is the observable

An unpolarized lithium beam would be one more light ion: its inclusive cross section is carried by  $F_1$  and  $F_2$  alone — for an unpolarized target the spin-1 cross section has exactly the spin- $\frac{1}{2}$  form [27] — and it would add an  $A = 6, 7$  point to the EMC and shadowing curves that  ${}^4\text{He}$  and  ${}^{12}\text{C}$  already map. That point is not without interest: the high- $x$  EMC ratios of  ${}^6\text{Li}$  and  ${}^7\text{Li}$  are in the target list of JLab E12-10-008 [28], motivated by the finding that the  ${}^9\text{Be}$  EMC effect follows the local density of its  $\alpha$  clusters [29], and the  $\alpha$ -tagged structure function of the triton or deuteron cluster versus spectator momentum would measure the  $\alpha$ - $t$  and  $\alpha$ - $d$  momentum distributions VMC predicts [13]. But every flagship item of the programme is a spin observable, and the two tensor ones need the tensor moment specifically:  $b_1$  is *defined* as the difference between the  $m = 0$  and  $m = \pm 1$  quark distributions, so averaging over  $m$  erases it, and  $\Delta$  needs both a target spin of at least 1 and a transverse alignment axis to define an azimuth (§4). The polarized EMC effect is a statement about  $g_1$ , and tagged DIS reads the spin of the cluster (§5). Polarized lithium would be the first polarized ion at a collider with a nuclear core — the first place where the spin structure of a *nucleus*, rather than of a nucleon, is studied at collider kinematics — and tensor polarization, on which both  $b_1$  and  $\Delta$  depend, is the scarcer resource: its survival through charge breeding and acceleration is unmeasured, and its polarimetry requires dedicated development [3,4].

### 3.2 · Fixed target or collider

Polarized lithium has existed for decades as a fixed target, but only in the solid state: dynamically polarized  ${}^6\text{LiD}$  was the deuteron target of SLAC E155 and of COMPASS,  ${}^7\text{LiH}$  has been polarized at Saclay (47% for  ${}^7\text{Li}$ , 56% for the protons) [30], and  ${}^7\text{LiD}$  is the material of the approved CLAS12 polarized-EMC experiment [17,20,31]. In these compounds the lithium polarizes along with the hydrogen isotope because all spin species share one spin temperature: COMPASS reached deuteron polarizations of  $+54\%/-47\%$ , whence  $51 \pm 3\%$  for  ${}^6\text{Li}$  and  $92 \pm 4\%$  for the  ${}^7\text{Li}$  admixture [31]; E155 measured  ${}^6\text{Li}$  at 97% of the deuteron's [17]; in  ${}^7\text{LiD}$  at 5 T the CLAS12 proposal expects 65–80% for  ${}^7\text{Li}$  [20]. Three limits are structural. (i) *Dilution*: only  $f \approx 0.35\text{--}0.45$  of the nucleons in  ${}^6\text{LiD}$  are polarizable once the helium coolant is counted — better than 0.22 for  $\text{ND}_3$  and 0.15 for  $\text{NH}_3$ , far from the 1.0 of a stored beam [17,31]. (ii) *Tensor polarization*: in thermal equilibrium it follows from the vector one,  $P_{zz} = 2 - \sqrt{4 - 3P_z^2}$ , so  $P_z = 0.5$  gives  $P_{zz} = 0.20$ ; radio-frequency "hole burning" reaches 29–37% in deuterated butanol at the cost of the vector polarization, and the JLab  $b_1$  experiment is conditionally approved on demonstrating 30% [32]. An atomic-beam source prepares the tensor moment directly: HERMES ran its gas target at  $P_{zz} = +0.89$  and  $-1.65$ , "a combination which is not possible for solid-state polarized targets" [33]. (iii) *Spectator loss*: a spectator  $\alpha$  of 100–300 MeV/c carries only 1.3–12 MeV in the target rest frame and stops within 10–250  $\mu\text{m}$  of lithium deuteride [34], inside cells 2–5 cm long at JLab and 30–60 cm at COMPASS, and a coherent  ${}^6\text{Li}$  recoil at  $|t| = 0.05 \text{ GeV}^2$  has only 4.5 MeV. The detectors that do tag slow spectators, BoNuS (protons from 70 MeV/c) and

ALERT ( $\alpha$  from 230 MeV/c), work with unpolarized gas at a few atmospheres [35,36]; polarized lithium outside dynamically polarized solids has never been used in lepton scattering.

The collider removes each limit at once. The beam is pure lithium, its polarization prepared atom by atom and measured in the ring; the spectator  $\alpha$  or triton carries the beam momentum per nucleon into the far-forward spectrometers at a few milliradians; the intact coherent recoil can be tagged down to  $p_T \approx 0$  at the secondary focus of a second interaction region [37]; and  $\sqrt{s_{NN}}$  up to 99.5 GeV for  $e(18 \text{ GeV}) \times {}^6\text{Li}(137.5 \text{ GeV/u})$  opens  $x$  down to  $\sim 10^{-4}$  at  $Q^2 \geq 1 \text{ GeV}^2$ , where JLab at 11 GeV stops at  $x \approx 0.06$  and HERMES stopped at 0.01 [20,33] — the region where gluons dominate, where HERMES's  $b_1$  was largest (§5.1), and where coherence holds. What stays with the fixed target is luminosity:  $10^{35} \text{ cm}^{-2} \text{ s}^{-1}$  (CLAS12 at 10–15 nA, Hall C at 115 nA) [20,32] against an electron–nucleon luminosity of  $10^{33}\text{--}10^{34} \text{ cm}^{-2} \text{ s}^{-1}$  at the EIC [1,2]. The high- $x$  polarized EMC effect and the high- $x$   $b_1$  of the deuteron are therefore rightly JLab measurements; the EIC takes the low- $x$  and high- $Q^2$  regions, the tensor observables at full polarization, and everything that needs a detected spectator or recoil (Table 1).

| MEASUREMENT                                                | AT A FIXED TARGET                                                                                                                                                                                                               | WHAT ONLY THE COLLIDER ADDS                                                                                                                                 |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Polarized EMC effect, ${}^7\text{Li}$ ( $g_1^A$ )          | Approved: JLab E12-14-001 (CLAS12, polarized ${}^7\text{LiD}$ , $0.06 < x < 0.8$ , $1 < Q^2 < 15 \text{ GeV}^2$ , $P_z \approx 0.8$ assumed) [20]                                                                               | $x < 0.06$ ; the $Q^2$ evolution that carries the gluon-spin EMC effect [22]; an undiluted beam; $\alpha$ -tagging of the struck triton and its virtuality  |
| Tensor $b_1$ , ${}^6\text{Li}$                             | Deuteron only: E12-13-011 ( $\text{ND}_3$ , $0.16 < x < 0.49$ , $P_{zz} \geq 0.3$ required) [32]; ${}^6\text{LiD}$ would give $P_{zz} \approx 0.2$ with the deuteron term to be subtracted, and no such measurement is proposed | $x$ down to a few $10^{-3}$ , below the $x \approx 0.01$ where HERMES found its largest $b_1$ (§5.1); pure-tensor fills at $P_{zz} \approx 0.6\text{--}0.8$ |
| Exotic glue $\Delta$ , ${}^6\text{Li}$ ( ${}^7\text{Li}$ ) | Letter of intent only: LOI12-16-006 ( ${}^{14}\text{N}$ in $\text{NH}_3$ , $x < 0.3$ , 2-12% alignment, in-plane spectrometers — $\cos 2\phi$ "difficult with a fixed target") [38]                                             | Transverse tensor polarization $\geq 0.6$ with full $2\pi$ azimuthal acceptance; the gluon-dominated region $x \lesssim 0.1$ and a $Q^2$ lever arm          |
| Spectator-tagged DIS ( $\alpha$ , $t$ )                    | Not feasible (§3.2, iii)                                                                                                                                                                                                        | Far-forward detection of $\alpha$ and $t$ at beam velocity; pole extrapolation to the free cluster [39,40]                                                  |
| Coherent intact- ${}^6\text{Li}$ channel                   | Not feasible: needs $x \lesssim 0.01$ and a 2-9 MeV recoil                                                                                                                                                                      | A near-beam tag of the intact recoil (§5.2)                                                                                                                 |

**Table 1** — what a fixed target covers and what only the collider adds, measurement by measurement.

## 4 • Formalism

### 4.1 • Deep inelastic scattering

An electron with four-momentum  $k$  scatters from a nucleus of momentum  $P$  by exchanging a virtual photon  $q = k - k'$ . The kinematic invariants, in the per-nucleon convention used throughout the programme (energies and structure functions per nucleon,  $x \leq 1$ ), are

$$Q^2 = -q^2, \quad x = \frac{Q^2}{2P \cdot q}, \quad y = \frac{P \cdot q}{P \cdot k}, \quad Q^2 = x y s, \quad W^2 = M^2 + Q^2 \frac{1-x}{x}$$

where  $x$  is the momentum fraction of the struck parton,  $y$  the energy-transfer fraction and  $W$  the invariant mass of the hadronic final state. The unpolarized cross section is carried by two structure functions,

$$\frac{d^2\sigma}{dx dQ^2} = \frac{4\pi\alpha^2}{x Q^4} \left[ \left(1 - y + \frac{y^2}{2}\right) F_2(x, Q^2) - \frac{y^2}{2} F_L(x, Q^2) \right], \quad F_1 = \frac{F_2}{2x(1+R)}$$

with  $R = \sigma_L/\sigma_T$ , for which the programme's projections use a placeholder of the right magnitude (0.14–0.18 at the kinematics of interest) and carry the SLAC/E143 R1998 world fit [41] behind a switch whose effect is quantified in §5.2. In the parton model  $F_2 = \sum_q e_q^2 x[q(x) + \bar{q}(x)]$ : the cross section measures quark distributions weighted by squared charge. For a nucleus we use  $F_2^A/A$ , so rates are quoted against luminosity per nucleon. Polarization introduces additional structure functions whose number and character depend on the target spin.

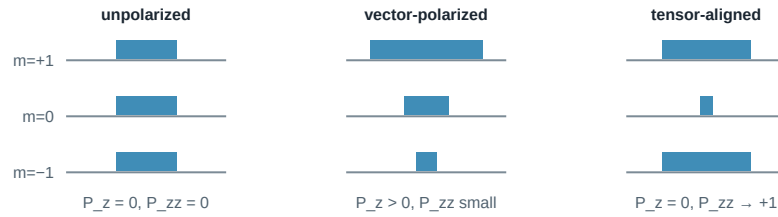
## 4.2 · Spin-1 targets

A spin- $\frac{1}{2}$  target contributes two spin structure functions,  $g_1$  and  $g_2$ . The principal observable is  $g_1$ , the helicity-weighted quark distribution, measured through the double-spin asymmetry  $A_{\parallel} \approx D \cdot g_1 / F_1$  ( $D$  is a kinematic depolarization factor; the  $\approx$  drops a target-mass term worth a factor  $1 + \gamma^2$ ,  $\gamma^2 = 4M^2 x^2 / Q^2 \leq 0.010$  rate-weighted in the window of Table 3) with both beams longitudinally polarized and the electron helicity reversed bunch by bunch.

A spin-1 target carries additional structure. Its spin state along a chosen axis is specified by three populations  $p_+$ ,  $p_0$ ,  $p_-$  of the projections  $m = +1, 0, -1$ , and two moments characterize the fill:

$$P_z = p_+ - p_- \in [-1, 1], \quad P_{zz} = p_+ + p_- - 2p_0 = 1 - 3p_0 \in [-2, 1]$$

$P_z$  is the familiar vector polarization.  $P_{zz}$ , the *tensor* or alignment polarization, is specific to spin  $\geq 1$ : it distinguishes spin along  $\pm z$  from spin in the transverse plane ( $m = 0$ ) and is nonzero even when equal numbers point up and down. An unpolarized beam has  $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$  and both moments vanish; a pure  $m = 0$  beam has  $P_z = 0$  but  $P_{zz} = -2$  (Figure 1).



**Figure 1** — populations of the three spin projections for an unpolarized, a vector-polarized and a tensor-aligned spin-1 fill. The tensor moment measures the  $m = \pm 1$  population against  $m = 0$  and is independent of the vector one.

Hoodbhoy, Jaffe and Manohar showed [27] that a spin-1 target has, beyond  $F_1$ ,  $F_2$ ,  $g_1$ ,  $g_2$ , four tensor-polarized structure functions  $b_1$ – $b_4$ , and Jaffe and Manohar identified one more, the double-helicity-flip function  $\Delta$  [42]. At leading twist the tensor sector reduces to  $b_1$  (with  $b_2 = 2x b_1$  as its Callan–Gross partner) and  $\Delta$ . For unpolarized electrons on a target whose spin axis makes angle  $\theta_m$  with the beam, the cross section for each pure  $m$  state is [27,42]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[ F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left( F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_m \Delta(x, Q^2) \cos 2\phi \right]$$

$$a_m = \frac{1}{4} c_m (3 \cos^2 \theta_m - 1), \quad c_m = 3|\lambda_m| - 2 \rightarrow (1, -2, 1) \text{ for } m = (+1, 0, -1)$$

where  $\phi$  is the azimuth of the lepton plane about the photon direction, measured from the transverse projection of the spin axis. We write the  $b_1$ ,  $b_2$  terms with the sign of [27] as transcribed in [5]; HERMES and Cosyn et al. [43] use the opposite sign ( $A_{zz} = -(2/3) b_1 / F_1$  for the axis along the photon). Only the sign of  $A_{zz}$  relative to  $b_1$  is affected; magnitudes, errors and the  $\Delta$  term are not.

The weights  $(1, -2, 1)$  organize all tensor observables: the  $m = \pm 1$  states enter together and against  $m = 0$ , precisely the combination measured by  $P_{zz}$ , and averaging over a fill replaces every  $c_m$  by  $P_{zz}$ . Two handles follow. With the spin axis *longitudinal*, the rate difference between fills gives  $A_{zz} \propto b_1$ ; with it *transverse* ( $\theta_m = 90^\circ$ ), the  $\phi$  distribution of the scattered electron acquires a  $\cos 2\phi$  modulation proportional to  $\Delta$ . Both require tensor polarization; neither requires polarized electrons.

$^7\text{Li}$ , with spin  $3/2$ , has four populations and three moments (vector, tensor/quadrupole, octupole). Its rank-0 and rank-1 sectors are exact analogues of the spin- $\frac{1}{2}$  case and host the polarized EMC measurement; its rank-2 sector is the formal analogue of the spin-1 tensor sector, for which no structure-function basis has been published.

## 5 · The observables

### 5.1 · The tensor structure function $b_1$

In parton language  $b_1$  compares quark distributions in an aligned versus a transverse nucleus [27],

$$b_1(x) = \frac{1}{2} \sum_q e_q^2 \left[ q^{(0)}(x) - \frac{q^{(+1)}(x) + q^{(-1)}(x)}{2} \right], \quad \int_0^1 b_1(x) dx = 0 \quad (\text{Close – Kumano, no tensor sea})$$

where  $q^{(m)}$  is the quark distribution in a target with spin projection  $m$ . For free spin- $\frac{1}{2}$  nucleons, or a pure S-wave bound state, all  $m$  states are equivalent and  $b_1 = 0$ , so a nonzero  $b_1$  measures directly how binding modifies the partonic content: D-wave admixture, pion exchange, hidden-colour components, or a tensor-polarized sea, which would violate the Close–Kumano sum rule [44].

The only existing measurement is HERMES on the deuteron [33]:  $A_{zz}$  up to  $2 \times 10^{-2}$  over  $0.01 < x < 0.45$  and  $b_1(x = 0.012) = 0.112 \pm 0.055 \pm 0.028$ , unexpectedly large at low  $x$ , with a zero crossing near  $x \approx 0.2$  and a sum-rule integral  $2\sigma$  from zero. Convolution calculations predict  $|b_1| < 10^{-3}$  at  $x \gtrsim 0.2$ , an order of magnitude below the data [45], whereas a model with pions and a hidden-colour probability of only 0.15% reproduces it [46]. Both curves are digitized into this programme's fast simulation, from Fig. 4 of [45] and Fig. 5 of [46]; over the  $x$  range the figures cover, the convolution respects the Close–Kumano sum rule,  $\int b_1 dx = 4.6 \times 10^{-4}$ , and the pionic model does not,  $5.9 \times 10^{-3}$ , as its tensor-polarized sea requires. JLab E12-13-011 will remeasure  $0.16 < x < 0.49$  with a solid  $\text{ND}_3$  target [32]; the EIC adds the low- $x$  region and the  $Q^2$  lever arm. The beam is delivered in three fills of different tensor polarization; with the estimator  $(N_+ + N_- - 2N_0)$  over matched luminosities all  $m$ -independent physics cancels [5]. At collider kinematics a clean extraction favours orienting the spin axis along the momentum transfer  $\mathbf{q}$  rather than along the beam [43].

## 5.2 · $\Delta(x, Q^2)$ : exotic glue

The  $\cos 2\phi$  term corresponds to the virtual photon flipping its helicity by two units between absorption and emission in the forward Compton amplitude. A spin- $\frac{1}{2}$  particle cannot absorb two units of helicity, so neither quarks nor bound nucleons generate  $\Delta$  at leading twist; gluons can.  $\Delta$  is what the gluon transversity distribution — an interference of gluon helicity states that exists only for  $J \geq 1$  — generates at order  $\alpha_s$  through a photon–gluon convolution [42,47], so a nonzero  $\Delta$  in a nucleus is direct evidence for glue not inherited from individual nucleons, hence "exotic glue". It has never been measured; the one prior experimental document is a letter of intent (Table 1).

The measurement uses unpolarized electrons on a tensor-polarized nucleus whose alignment axis lies in the transverse plane, and fits the azimuthal distribution about that axis. Averaged over the fill, the amplitude and its inversion back to  $\Delta$  are

$$A^{\cos 2\phi} = -\frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2$$

Writing  $F_2 = 2x(1+R)F_1$  removes both structure functions from the ratio  $\Delta/F_1$  except through  $R$ , which is the one structure-function input that reaches the physics result —  $F_2$  cancels exactly. At the four kinematic bins of the companion projection ( $x = 0.011\text{--}0.14$ ,  $Q^2 = 1.1\text{--}14 \text{ GeV}^2$ ), replacing the placeholder  $R$  by the R1998 fit moves  $\Delta/F_1$  by +18, +19, +8 and -4% [5].

The expected size spans per mille to per cent, and the only absolute normalizations available are moment estimates: a bag-model calculation for the spin- $3/2$   $\Delta^{++}$  baryon,  $\int x \Delta dx = -0.012 \alpha_s$  [47], and lattice-QCD calculations of gluon transversity in the  $\phi$  meson (clearly nonzero, at an unphysical pion mass) [48], with the analogous matrix element roughly ten times suppressed in the deuteron [49]. Percent-level asymmetries appear only in proton–deuteron Drell–Yan under the maximal assumption  $\Delta_{Tg} = \Delta g$ , which its authors call optimistic [50]. No calculation exists for  ${}^6\text{Li}$  or any  $A > 2$  nucleus, and the literature brackets the nuclear  $\Delta$  both ways: suppressed, as in the lattice deuteron, or enhanced by binding in analogy with the EMC effect [3]. The projection of [5] transfers the bag moment to  ${}^6\text{Li}$  as a scenario — a  $\Delta$  whose  $x$ -moment equals the bag value — and one EIC year then measures the amplitude at  $21\text{--}44\sigma$  per bin (Table 3).

**The coherent channel.** The same modulation can be measured in coherent diffraction,  $e {}^6\text{Li} \rightarrow e' X {}^6\text{Li}(\text{g.s.})$ , where the nucleus stays in its ground state and recoils intact. There the modulation mixes two mechanisms: the nuclear *shape*, a deformed gluon density modulating the diffractive slope (computed for polarized deuterons in the only existing calculation of this kind [51]), and exotic glue proper, which persists at small  $x$ , where it connects to the elliptic-gluon programme [52].  ${}^6\text{Li}$  discriminates between them because its quadrupole moment is anomalously small (§2.2): the shape term is bounded and linear in  $t$ , so a flat-in- $t$  modulation cannot be a shape effect. Detecting the intact recoil is the difficulty. The nucleus



keeps the beam's rigidity and emerges within a milliradian or two of the beam, so the only handle is its angle against the  $10\sigma$  beam envelope and the Roman-Pot aperture, and the tagged fraction is an exponential in the square of the beam divergence,  $\exp[-B(10\sigma_\theta \cdot A \cdot p_u)^2]$ , with  $B \approx 50 \text{ GeV}^{-2}$  the slope of the coherent  $t$  distribution,  $\sigma_\theta$  the beam angular divergence and  $p_u$  the momentum per nucleon. With the Yellow Report's divergences the fraction is  $5 \times 10^{-7}$ ,  $8 \times 10^{-26}$  and  $4 \times 10^{-13}$  at the three  ${}^6\text{Li}$  configurations for a circular envelope, and the real anisotropic rectangle with the measured pot edge ( $2.50 / 1.51 / 0.53 \text{ mrad}$ , re-measured on 2026-08-28 in the current ePIC geometry, the  $5 \times 41$  figure conditional on which of that ring setting's two lattices a  ${}^6\text{Li}$  fill runs in and worth  $\times 0.64$  in the other) as a second constraint per axis lowers it further, to  $9 \times 10^{-10}$ ,  $6 \times 10^{-27}$  and  $7 \times 10^{-14}$  [5,26]: the channel does not exist at IP6 with any published optics, though which of the two constraints delivers that verdict has changed hands with the re-measurement — the silicon at  $5 \times 40.8$ , where  $2.50 \text{ mrad}$  exceeds a  $2.20 \text{ mrad}$  envelope, and the beam at  $18 \times 137.5$ , where the aperture halved to  $0.53 \text{ mrad}$  against an envelope of  $0.92$ . It requires either a dedicated tagging optics —  $\beta^*$  raised so that the envelope sits at  $p_T = 1/\sqrt{B} \approx 0.14 \text{ GeV}$ , where the tagged fraction is  $1/e$  (a factor  $15\text{--}60$  in  $\beta^*$  for a circular envelope), with pots that follow — or the secondary focus of the second interaction region, which reaches  $p_T \approx 0$  [37]. The tagging optics recovers a  $32\text{--}42\%$  tag with the horizontal  $\beta^*$  raised  $50\text{--}176\times$  and the vertical plane untouched, at a factor  $7\text{--}13$  in luminosity, which makes it an  $8\text{--}11\sigma$ -per-year measurement of the shape term and a three-to-four-year one of a  $1\%$  exotic-gluon term,  $3\text{--}8\times$  below what a second interaction region's secondary focus would give at the same luminosity [5]; the projections of [5] assume a  $0.20 \text{ GeV}$  envelope,  $13.5\%$  of the coherent yield, at the high-acceptance luminosity, as the reference tag.

### 5.3 · The polarized EMC effect: $g_1$ of a bound proton

The EMC effect — the  $\sim 10\text{--}20\%$  suppression of  $F_2$  per nucleon in a nucleus at  $0.3 \lesssim x \lesssim 0.7$  — has been measured for forty years without a settled explanation. Its spin analogue has never been measured: does the *spin* structure of a bound nucleon change more, less, or the same as its momentum structure? The observable is the polarized EMC ratio

$$\Delta R_A(x) = \frac{g_1^A(x)}{P_p g_1^p(x) + P_n g_1^n(x)}$$

where  $P_p$ ,  $P_n$  are the effective nucleon polarizations from nuclear structure. With the effective polarizations of §2.2,  ${}^7\text{Li}$  is to a good approximation one polarized proton plus six spectators, with computable corrections. Two classes of model differ sharply: Cloët, Bentz and Thomas (mean field, NJL) obtain a polarized EMC effect about twice the unpolarized one [21], Tronchin, Matevosyan and Thomas equal effects [53]. Both calculations are digitized here from their published figures, which sharpens the comparison and moves it. Within Cloët, Bentz and Thomas's own  ${}^7\text{Li}$  calculation — their Eq. (23)  $R^{3/2,3/2}$ , the curve plans/01 defines  $\Delta R_A$  on, with their Eq. (26)  $R^{(3/2,1)}$  digitized alongside it — the ratio of effects  $(1 - R_{\text{pol}})/(1 - R_{\text{unpol}})$  is  $2.25, 1.69, 1.41$  and  $1.14$  at  $x = 0.40, 0.45, 0.50$  and  $0.60$ , falling to a minimum of  $1.06$  at  $x \approx 0.70$  without reaching unity, against  $1.01, 0.98, 1.00$  and  $1.08$  for Tronchin, Matevosyan and Thomas: the factor two is a statement about  $x \approx 0.4$  rather than a constant, and the two converge at the top of the valence region instead of separating there. The two are computed for different targets and scales —  ${}^7\text{Li}$  at  $Q^2 = 5 \text{ GeV}^2$  and isospin-symmetric nuclear matter at  $Q^2 = 10 \text{ GeV}^2$  — so a projection needs them on one common  ${}^7\text{Li}$  unpolarized baseline, matched here in the strength of the unpolarized effect over  $0.35 < x < 0.65$ ; carried across that way their predictions for  $\Delta R$  differ by  $0.044, 0.040, 0.034$  and  $0.059$  at  $x = 0.09, 0.28, 0.45$  and  $0.71$ . That transfer factor is a single constant fitted in the valence window and then applied at every  $x$ , so only the part of the separation that falls inside the window is a statement about the two calculations. Below it they agree: the two published polarized ratios differ by less than  $0.008$  across the whole decade  $0.028 < x < 0.30$  and by  $0.002$  at  $x = 0.09$ , and  ${}^7\text{Li}$ 's unpolarized effect there is an enhancement rather than a depletion, so a strength ratio has nothing to scale — the near-constant  $0.04$  the projection plots at low  $x$  is the rescaling, not the papers. Inside the window the comparison is the one the two camps actually make, and it fades across it: the transferred nuclear-matter depletion tracks  ${}^7\text{Li}$ 's own unpolarized  $0.034, 0.048$  and  $0.087$  at  $x = 0.40, 0.45$  and  $0.65$  to within  $0.005$ , which is what "polarized  $\approx$  unpolarized" means, against  $0.077, 0.082$  and  $0.094$  for the mean-field calculation, so  $\Delta R$  separates by  $0.040$  at  $x = 0.36$  and by only  $0.011$  at  $0.65$  as the ratio of effects falls to unity. The discriminating region is therefore  $x \approx 0.35\text{--}0.45$  — inside the valence region, as before the digitization, but at its lower edge and with half the separation the constants  $2$  and  $1$  implied. A second effect points the same way: the three-flavour  $g_1$  over five-flavour  $F_2$  scheme of §4.2 costs  $0.3\%$  of  $F_2^A$  at  $x = 0.35$  falling to  $0.03\%$  at  $0.65$ , against  $2.6\%$  at  $x = 0.09$ , where it is as large as the separation drawn. The two

map onto the competing mean-field and short-range-correlation pictures of the EMC effect itself. Unique to the EIC is the  $Q^2$  evolution of  $g_1$ , driven by the polarized gluon distribution; the predicted *gluon-spin* EMC effect is larger still, with  ${}^7\text{Li}$  the most promising case [22], and the collider measures it free of target-fragment absorption. The measurement uses the longitudinal double-spin configuration of §4.2, i.e. the vector polarization the source delivers most readily.  ${}^6\text{Li}$  and the deuteron supply the isoscalar companions, so the three together separate the u- and d-quark spin modification — an inference of this programme, not yet in the literature.

### 5.4 • Spectator-tagged DIS

Detecting the  $\alpha$  spectator, which carries beam velocity into the far-forward spectrometers, identifies the struck object as the deuteron cluster of  ${}^6\text{Li}$  or the triton cluster of  ${}^7\text{Li}$ , so the beam acts as a source of polarized light clusters at controlled virtuality. The framework is spectator tagging [39]: the spectator momentum is measured and extrapolated in its transverse component to the free-cluster pole, recovering free-particle structure with binding effects removed by construction rather than modelled. Tagged *tensor* asymmetries reach  $O(1)$  at spectator momenta of  $\sim 300$  MeV/c, where the D-wave dominates [54], far larger than the inclusive asymmetries. Cosyn and Weiss give that asymmetry in closed form and this programme's generator reproduces it: at fixed spectator momentum the wave-function tensor asymmetry factorizes into their  $P_2(\cos \theta_k)$  angular factor to better than  $10^{-5}$  in every angular cell away from the zero of  $P_2$  at  $\cos \theta_k = 1/\sqrt{3}$ , where the ratio is unbounded, its radial envelope peaks at  $k = 0.310$  GeV/c against their 0.30 GeV/c for the AV18 wave function, and under the normalization  $A_{T\parallel} = -2 A_{zz}^{wf}$  its extremes extrapolate to +1.000 at  $\theta_k = 90^\circ$  and  $-2.000$  at  $\theta_k = 0$ , the two values they tabulate. The benchmark is the  $e^+{}^3\text{He}$  double-tagging study of Frišćić et al. [55], where tagging improved on effective-polarization unfolding by a factor 2–3, and 10 at low  $x$ .

Where each fragment is detected follows from magnetic rigidity alone: a beam-velocity fragment bends in the hadron optics according to the ratio of mass-to-charge ratios  $R = (m/Z)_{\text{frag}}/(m/Z)_{\text{beam}}$  (Table 2). The  $(A/Z)$  ratio approximates it to  $5 \times 10^{-3}$ ; the  $\alpha$  is more bound per nucleon than the lithium it leaves, so it sits just below beam rigidity rather than on it.

| BEAM                            | TAG<br>FRAGMENT | R                    | DESTINATION AT IP6                                                                         |
|---------------------------------|-----------------|----------------------|--------------------------------------------------------------------------------------------|
| ${}^7\text{Li}$ ( $A/Z = 7/3$ ) | $\alpha$        | 0.85571              | Roman-Pot main window ( $R \in 0.60\text{--}0.95$ ) — works now                            |
| ${}^7\text{Li}$                 | t               | 1.28971              | over-rigid: Roman-Pot inner side + ZDC (measured; see text)                                |
| ${}^6\text{Li}$ ( $A/Z = 2$ )   | $\alpha / d$    | 0.99813 /<br>1.00452 | within $5 \times 10^{-3}$ of beam rigidity: inside the beam envelope except the $p_T$ tail |
| ${}^6\text{Li}$                 | p / n           | 0.50251 / –          | Off-Momentum Detectors / ZDC (veto handles)                                                |

**Table 2** — far-forward routing of the tag fragments at IP6, from the rigidity ratio  $R$  alone; the windows are those of Jentsch, Tu and Weiss [40]. The over-rigid triton was counted as lost until 2026-08-28, on the argument that the rigidity windows carry no  $R > 1$  branch; that is now settled by measurement rather than assumption. A full simulation of the current ePIC geometry puts a triton on the Roman-Pot silicon in 60 of 60 events at every configuration — the gun was fired at  $R = 1.286$  against the physical-mass 1.28971 of the table, a 0.3% rigidity difference worth under 0.3 mm at the pot plane —, on the inner side of the bend at  $dx = +66$  mm, with a zero-degree-calorimeter deposit behind it in 80–98% of the same events, so the  ${}^7\text{Li}$   $\alpha + t$  double tag is available at IP6 as an RP-inner track plus calorimeter energy [5,26]. What remains unchecked is a reconstruction: the hit is at  $y \approx 0$  and needs no  $p_T$ , but the fast simulation routes in angle and cannot see the per-band vertical insertions that put a real acceptance hole between  $\theta_x = -1.55$  and  $-2.53$  mrad.

In the cluster model of [5], the  ${}^7\text{Li}$   $\alpha$  tag reaches the Roman Pots with  $\approx 97\%$  acceptance at every configuration and every optics (95–99% over the wave-function tail scale  $\beta = 0.2\text{--}0.4$  GeV), because  $R = 0.856$  puts it in the main Roman-Pot window and no beam envelope is in its way. That routing is in angle alone, and at  $5 \times 40.8$  it is optimistic: the per-energy insertion holds the silicon band off to  $|y| \geq 18.0$  mm there, so an  $\alpha$  at  $y \approx 0$  sees no silicon at station 1 at all and the full simulation of the current geometry finds it at station 2 only, on a module edge, in 57% of events (Report 3 [26] Table 6). At  $10 \times 99.5$  and  $18 \times 117.9$  the insertion is nearer and the angle-only 97% stands. The  ${}^6\text{Li}$   $\alpha$  is the opposite case. It sits at  $R = 0.998$ , so it can be tagged only through the  $p_T$  tail above the near-beam cut, and whether that tail exists is an optics question. At the Yellow Report high-acceptance optics it does not: the tail lies inside the  $10\sigma$  envelope at every configuration, and the 1.9 / 1.7 / 2.6% the tag reads at 40.8 / 99.5 / 137.5 GeV/u is the off-rigidity slice the Roman-Pot



window catches together with the over-rigid inner branch, not a tail measurement at all. The tagging optics of §5.2 reopens the tail and recovers 35 / 28 / 29% at 1/7–1/13 of the luminosity, the same trade and the same request as the coherent channel, and Report 3 [26] Table 6 tabulates both tags against both optics. The triton of  ${}^7\text{Li}$  is the one tag the re-measurement transformed: routed as lost while  $R = 1.29$  had no window, it is tagged in 78 / 92 / 94% once the measured over-rigid branch is applied, almost all of it on the inner side of the Roman Pots and the rest in the B0 at  $5 \times 40.8$ , where the angles are largest (Table 2). Because the  ${}^6\text{Li}$  tag under the tagging optics is a  $p_T$ -tail measurement, its model uncertainty there is one-sided upward: the cluster model undershoots the official BeAGLE e+d sample by 2.5 $\times$ , 6.9 $\times$  and 28 $\times$  at  $p_T > 0.2, 0.3$  and  $0.45$  GeV. The tagging programme therefore begins with  ${}^7\text{Li}$  at IP6, while  ${}^6\text{Li}$   $\alpha$ -tagging — and the coherent channel, which faces the same rigidity problem — motivates the near-beam tagging capability of §5.2.

## 6 • Requirements and reach

All projections use luminosity per nucleon: 10 fb $^{-1}$ /u — 30 weeks at 10 $^{33}$  cm $^{-2}$  s $^{-1}$  with 60% efficiency in the Yellow Report's accounting [2], "one EIC year" — and 100 fb $^{-1}$ /u for the ten-year programme, at placeholder in-ring polarizations  $P_z = 0.7$ ,  $P_{zz} = 0.6$  and  $P_e = 0.7$ ; transport of the source polarization to the ring is an open accelerator question, and every statistical error scales as 1/(P $\sqrt{L}$ ), so halving  $P_{zz}$  costs a factor four in running time. The observables do not share a configuration: they demand different spin states, different far-forward optics — the coherent channel of §5.2 wants the tagging optics and pays its luminosity penalty for as long as it runs — and, between the polarized EMC effect and the tensor sector, different isotopes,  ${}^6\text{Li}$  and  ${}^7\text{Li}$  being separate fills and separate runs, and each projection assumes the full luminosity of the year in its own. The reaches of Table 3 are therefore per observable and cannot be added into one year; how the year is divided between them is a programme decision, priced in plans/07 WP2, under which every statistical error grows as one over the square root of the share while a luminosity quoted as a reach is unchanged by it. The statistical uncertainties behind every projection are

$$\delta A_{\parallel} = \frac{1}{P_e P_z \sqrt{N}}, \quad \delta A_{zz} = \frac{\sqrt{2/N}}{P_{zz}}, \quad \delta A^{\cos 2\phi} = \frac{\sqrt{2/N}}{P_{zz}} \text{ (single state)}, \quad \delta \hat{A} = \frac{2}{P_+ - P_0} \sqrt{\frac{2}{N}} \text{ (two states)}$$

The factor  $\sqrt{2}$  in the tensor uncertainties follows from  $\langle \cos^2 2\phi \rangle = \frac{1}{2}$  (equivalently, from the three-fill estimator variance); the polarization sits in the denominator because the fill dilutes the physical asymmetry. These formulas, and their behaviour under realistic  $\phi$ -acceptance holes (where a binned fit replaces the naive moment), are validated by the pseudo-experiments of [5]. Two of the four highest-significance  $\Delta$  bins sit at  $y \approx 0.01$  and the other two at  $y \approx 0.026$ , where the electron alone gives  $\delta y/y \approx 1.1$ –1.2 and 0.46 and the hadronic  $\Sigma$  method, with a PYTHIA 8 final state and 50 MeV of calorimeter noise, 0.32 / 0.22 / 0.29 / 0.15 at the mid configuration — a resolution set by the noise floor (0.2  $\rightarrow$  1.0 at  $y = 0.005$  for 0  $\rightarrow$  100 MeV) on a hadronic  $E - p_z$  sum of 0.2–0.5 GeV that no published ePIC study covers. Tensor observables have a structural advantage when the  $m = \pm 1$ -rich and  $m = 0$ -rich states are measured with the same detector and compared bin by bin. In their spin-state ratio the  $\phi$  acceptance cancels and relative-luminosity errors enter only as a constant, whereas a single-state fit also measures the detector's own  $\cos 2\phi$  modulation divided by  $P_{zz}$ . The two-state pattern  $P_+ = +0.6$ ,  $P_0 = -1.2$  is moreover 1.5 $\times$  more precise than a single state, provided the source delivers the  $m = 0$ -rich bunches at the same purity. The cancellation requires bunch-by-bunch alternation: a 10 $^{-3}$  difference of the acceptance harmonic between the states fakes half a 10 $^{-3}$  signal. The polarimetry scale and the tensor radiative corrections do not cancel. Table 3 collects, observable by observable, what the beam must provide and what the simulations project.

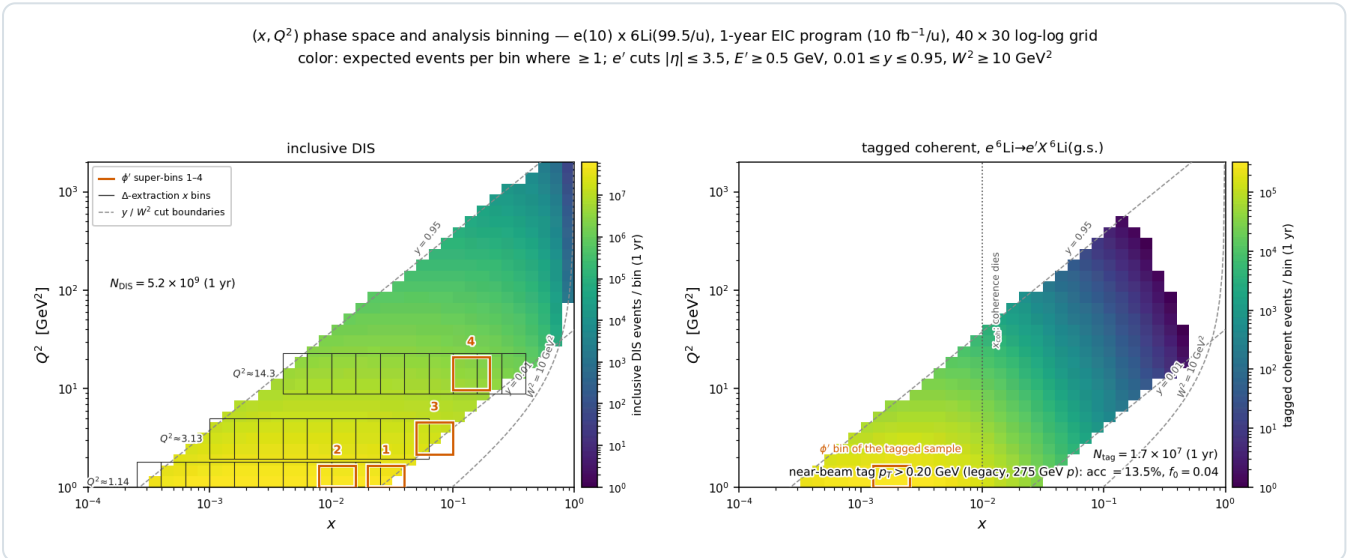
| OBSERVABLE                                                                          | BEAM POLARIZATION                                                                                                                                                                                                   | ELECTRONS                                                      | REACH AT 10 FB $^{-1}$ /U                                                                                                                                                                                                                                                                                                                                                     | BINDING REQUIREMENT                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $g_1^A$ , polarized EMC ( ${}^7\text{Li}$ ; ${}^6\text{Li}$ for the isoscalar pair) | Vector $P_z \geq 0.7$ , <b>longitudinal</b> (spin rotators). A spin-3/2 spin-temperature fill at $P_z = 0.8$ also carries $P_{zz} \approx 0.55$ and $P_{zzz} \approx 0.11$ ; the octupole term enters the asymmetry | Polarized, $P_e \approx 0.7$ , helicity flipped bunch by bunch | $\delta \Delta R \approx 4.2\%$ per x-bin at $x = 0.09$ , 4.1% at 0.28, 6.0% at 0.45 and 18.9% at 0.71 (three energies combined, on CT18NLO + NNPDFpol11 grids; 4.8, 5.1, 6.2 and 12.4% on the toy inputs the figure draws). Against the digitized curves (§5.3) the CBT-TMT difference is 1.04, 0.98, 0.57 and 0.31 $\sigma$ per bin there on the grid inputs, and 3.3, 3.1, | Polarimetry scale $\delta P_z/P_z \leq 3\%$ ; $P_p$ from VMC [13]; a $Q^2$ lever arm at fixed $x$ (gluon-spin EMC); the target-mass term dropped from $A_{\parallel} = D \cdot g_1/F_1$ biases $\Delta R$ <i>high</i> by 0.12 / 0.44 / 0.71 / 1.06% at $x = 0.09$ / 0.28 / 0.45 / 0.71 (inverse-variance weighted |

|                                                                         |                                                                                                                                                                                                                                                    |                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                         | numerator and is corrected through the fill model [20]                                                                                                                                                                                             |                                 | 1.8 and $1.0\sigma$ with $100 \text{ fb}^{-1}/\text{u}$ . Read where the two calculations actually differ — the valence window $0.35 < x < 0.65$ in which the transfer of §5.3 is defined — the best bin is $x = 0.36$ at $0.84\sigma$ at $10 \text{ fb}^{-1}/\text{u}$ and $2.7\sigma$ at 100 ( $0.72$ and $2.3\sigma$ on the toy inputs the figure draws); the larger figures the same projection returns at $x \approx 0.09$ – $0.14$ are carried by the transfer rather than by the papers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | over $Q^2$ and the three energies — the same weights as the $\delta\Delta R$ of the reach column), i.e. $2.8$ – $25\times$ below the polarimetry scale and $8.7$ – $40\times$ below the toy-input $\delta\Delta R$ the bound was computed against, $8.5$ – $36\times$ below the grid-input $\delta\Delta R$ the reach column leads with                                                             |
| $b_1$ via $A_{zz}$ ( ${}^6\text{Li}$ )                                  | <b>Tensor</b> $P_{zz}$ , longitudinal or along $\mathbf{q}$ [43]; fills of different $P_{zz}$ ("tensor thirds"), i.e. $m = 0$ bunches from the source                                                                                              | Unpolarized (helicity-averaged) | $\delta A_{zz} \approx 0.9 \times 10^{-4}$ per $x$ -bin (combined over $Q^2$ and the three energies) at $x < 0.05$ ( $0.7 \times 10^{-4}$ at $P_{zz} = 0.8$ ), rising to $\approx 10^{-3}$ at $x \approx 0.6$ , against the digitized $b_1$ of the embedded deuteron [46], carried into ${}^6\text{Li}$ by the per-deuteron→per-nucleon $\frac{1}{2}$ , the rank-2 tensor transfer $0.92$ and the $\frac{1}{3}$ dilution — $0.15$ in all, or $0.31$ applied to a $b_1$ already quoted per nucleon (not the vector $0.87$ , which is the wrong rank for a tensor structure function): $ A_{zz}  = 2.4 \times 10^{-4}$ at $x = 0.005$ rising to $1.4 \times 10^{-3}$ at $x = 0.07$ , i.e. $1.7$ , $4.8$ , $7.6$ and $10.9\sigma$ per bin at $x = 0.004$ , $0.009$ , $0.028$ and $0.071$ at $P_{zz} = 0.6$ , while the convolution [45] stays below $0.2\sigma$ everywhere: the two calculations are not a factor ten apart but the difference between a measurement and none. $x > 0.3$ is JLab's domain | Relative luminosity between fills at the $10^{-4}$ level (a $10^{-4}$ offset biases $A_{zz}$ by $1 \times 10^{-4}$ ; the luminosity-corrected estimator removes it exactly); tensor polarimetry, which EPIOS flags as requiring dedicated development (Lamb-shift or Breit-Rabi-type polarimeters) [3]                                                                                              |
| $\Delta$ via $\cos 2\phi$ ( ${}^6\text{Li}$ ; ${}^7\text{Li}$ eligible) | <b>Tensor</b> $P_{zz}$ with the alignment axis <b>transverse</b> (the native vertical direction); fills of opposite $P_{zz}$ sign flip the modulation — the decisive systematic check                                                              | Unpolarized                     | $\delta A = (1.4\text{--}4.5) \times 10^{-4}$ per bin in the four highest-significance bins ( $x = 0.011$ – $0.14$ , $Q^2 = 1.1$ – $14 \text{ GeV}^2$ ) against bag-moment-constrained amplitudes of $(0.44\text{--}0.95) \times 10^{-2}$ , i.e. $21$ – $44\sigma$ ; a $3 \times 10^{-3}$ flat amplitude stays $\geq 3\sigma$ per bin even at $P_{zz} = 0.3$ . Reconstructed, with the two-state ratio and a hadronic-method $y$ : $\delta \tilde{A} = (0.9\text{--}3.0) \times 10^{-4}$ ; with a PYTHIA 8 final state through the hadron-side response and the hadronic scale calibrated, the bin purities are $0.56$ – $0.75$ and the reconstructed-bin amplitude within $6\%$ of the true-bin value ( $0.42$ – $0.68$ and $5$ – $21\%$ low uncalibrated)                                                                                                                                                                                                                                            | Spin-state ratio of $m = \pm 1$ -rich and $m = 0$ -rich bunches alternating bunch by bunch; a hadronic-method $y$ with $\delta y/y \leq 0.3$ at the $y \approx 0.01$ – $0.03$ bins, i.e. a calorimeter noise floor $\leq 50 \text{ MeV}$ on a hadronic $E - p_z$ sum of $0.2$ – $0.5 \text{ GeV}$ (text); $\delta P_{zz}/P_{zz} \approx 3\%$ scale; tensor radiative corrections not yet calculated |
| Spectator-tagged DIS                                                    | ${}^7\text{Li}$ $\alpha$ -tag: any polarization for tagged $F_2$ ; vector $P_z$ with polarized electrons for tagged $g_1$ of the bound proton. ${}^6\text{Li}$ $\alpha$ -tag: tensor $P_{zz}$ for the order-unity tagged tensor asymmetries (§5.4) | As for the inclusive observable | ${}^7\text{Li}$ $\alpha \rightarrow$ Roman Pots, $\approx 97\%$ , optics-robust in angle (at $5 \times 40.8$ the measured per-band insertion leaves $57\%$ , station 2 only); ${}^7\text{Li}$ $t \rightarrow$ Roman-Pot inner side + ZDC, $78$ – $94\%$ ; ${}^6\text{Li}$ $\alpha$ beam-blind at IP6, $1.7$ – $2.6\%$ at the Yellow Report optics of the three configurations and $28$ – $35\%$ under the tagging optics of §5.2, at $1/7$ – $1/13$ of the luminosity (§5.4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Spectator $p_T$ resolution $\sim 10\%$ at $100 \text{ MeV}/c$ [40]; charge identification of $\alpha$ against $A/Z = 2$ fragments in the Roman Pots; pole extrapolation in $p_T$ [39]; the one-sided $p_T$ -tail model uncertainty                                                                                                                                                                  |

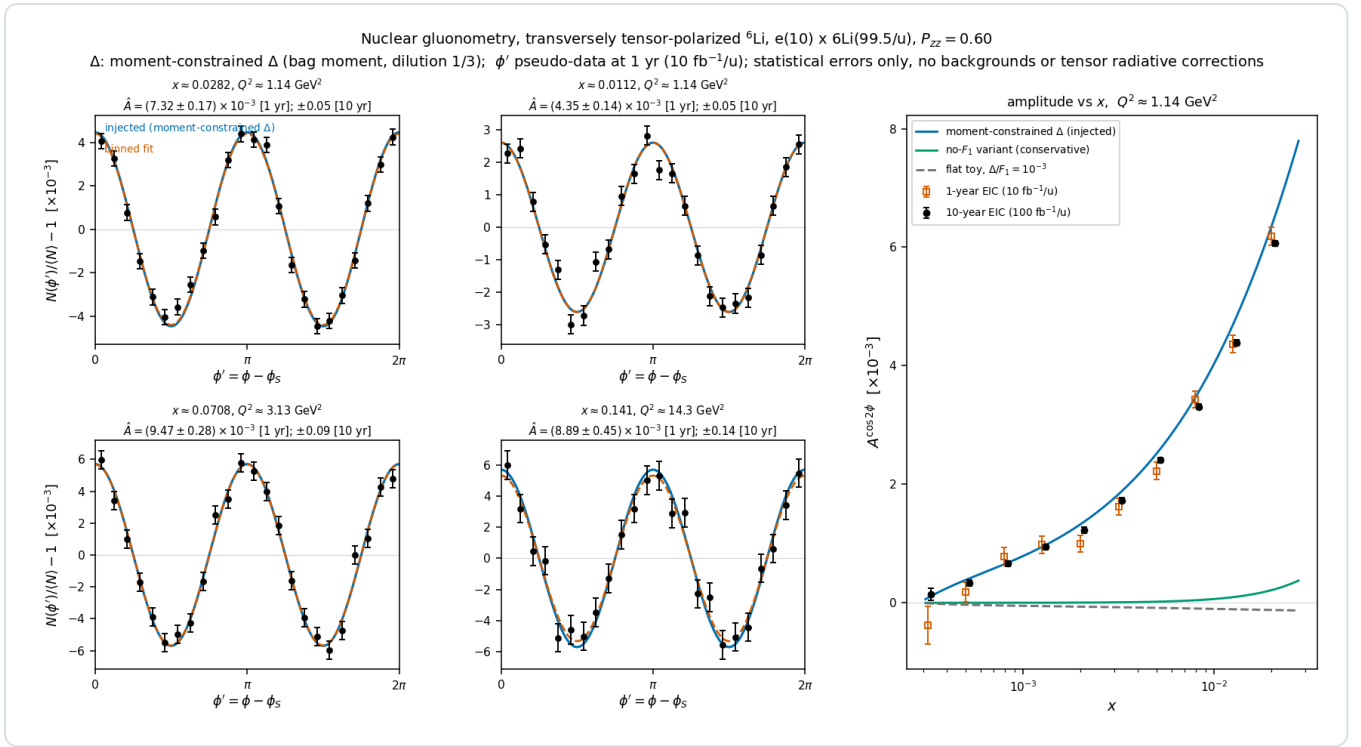
|                                              |                                                     |             |                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                 |
|----------------------------------------------|-----------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Coherent intact- ${}^6\text{Li}$ cos $2\phi$ | <b>Tensor</b> $P_{zz} \geq 0.6$ , <b>transverse</b> | Unpolarized | $x \leq 0.01$ , $ t  < 0.2 \text{ GeV}^2$ . For a tag with a 0.20 GeV near-beam envelope (13.5% of the coherent yield): $1.7 \times 10^7$ tagged events, best-bin $\delta A = 1.8 \times 10^{-3}$ ( $5\sigma$ at a 0.9% modulation), against a shape term $\langle a_2 \rangle = 0.036$ [0.018–0.059] scaled from the polarized-deuteron calculation [51] and a flat gluonic $3 \times 10^{-3}$ – $10^{-2}$ [5] | A near-beam tag (§5.2); Z-identification against $\alpha+d$ , both at beam rigidity; B0 calorimeter veto of the 3.56 MeV de-excitation $\gamma$ |
|----------------------------------------------|-----------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

**Table 3** — beam requirements and one-year reach by observable, from the programme's fast simulation and event-generator pseudo-experiments [5]. The  $\Delta$  and coherent rows are at  $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV/u})$ ; the  $b_1$  and polarized-EMC rows combine the three beam configurations; the  ${}^6\text{Li}$  tagging acceptances span the three configurations (40.8 / 99.5 / 137.5 GeV/u) and the  ${}^7\text{Li}$  ones are at 117.9 GeV/u. All structure-function inputs are scenarios with explicit bands.

Species by species:  ${}^6\text{Li}$  is the only spin-1 nucleus beyond the deuteron the EIC could polarize —  $b_1$  and  $\Delta$  do not exist for protons or  ${}^3\text{He}$  at all — and the lattice indication that the deuteron's exotic glue is suppressed [49] makes a bound nucleus the likelier place for a discovery [3].  ${}^7\text{Li}$  is the only bound polarized proton within reach of any collider, and the only light-ion beam whose spectator is a spin-0, tightly bound nucleus landing in the Roman-Pot window today. Figures 2 and 3 show the phase space and the projected  $\Delta$  measurement that Table 3 summarizes.



**Figure 2** — expected events per  $(x, Q^2)$  bin in one EIC year at  $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV/u})$ . Left: inclusive DIS ( $5.2 \times 10^9$  accepted events) with the bins of the cos  $2\phi$  analysis overlaid; the measurement is limited by polarization, acceptance and systematics, not by statistics. Right: the coherent intact- ${}^6\text{Li}$  channel for a tag with a 0.20 GeV near-beam envelope ( $1.7 \times 10^7$  tagged events; the panel's "HA optics" label is the proton optics the cut was first taken from — no published  ${}^6\text{Li}$  optics provides it, §5.2), confined to  $x \leq 0.03$ : above  $x \approx 0.01$  the coherence length  $\approx 0.1 \text{ fm}/x$  drops below the nuclear diameter ( $\approx 5 \text{ fm}$ ) and the coherent fraction falls off.



**Figure 3** — projected extraction of  $\Delta \cdot \cos 2\phi'$  pseudo-data (one EIC year,  $P_{zz} = 0.6$ ) in the four highest-significance ( $x$ ,  $Q^2$ ) bins, with the injected moment-constrained  $\Delta$  model and the binned fit; right, the extracted amplitude versus  $x$  against alternative model readings. The precision distinguishes between the models rather than only from zero. The reconstructed-level version uses the spin-state ratio in reconstructed bins [5].

## 7 • Summary

Lithium is the practical polarized nucleus for the EIC because three things coincide: an alkali atom that a single laser polarizes and an existing EBIS charge-breeds; two stable isotopes whose nuclear spin sits on a light cluster — a polarized deuteron in  ${}^6\text{Li}$ , a polarized proton in  ${}^7\text{Li}$  — bound to an inert  $\alpha$  that exact few-body theory describes; and ions whose spin dynamics in the ring are deuteron-like. Its physics is inaccessible without polarization, since  $b_1$ ,  $\Delta$ ,  $g_1^A$  and the tagged tensor asymmetries all vanish or are undefined without it, and largely inaccessible at a fixed target, where dilution, spin-temperature-limited tensor polarization and the loss of slow spectators are structural. Four measurements follow, each a first:  $b_1$  of a nucleus beyond the deuteron, the never-measured double-helicity-flip structure function  $\Delta$ , the polarized EMC effect on a bound proton at collider kinematics, and  $\alpha$ -tagged DIS on a polarized cluster. The requirements they place on the machine are specific — tensor polarization delivered as  $m = \pm 1$ -rich and  $m = 0$ -rich bunches alternating bunch by bunch, transverse polarization at the interaction point, relative luminosity between fills controlled to  $10^{-4}$ , and a near-beam tagging capability for  $A/Z = 2$  recoils that no published optics provides — and each is a concrete question to the accelerator and detector groups. The projections in the companion report [5] quantify what answering them buys.

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## Appendix A • Revision history

| DATE       | REVISION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-08-29 | Verification pass over the round. Table 3's reconstruction row restated on the 2026-08-28 toolchain review — purities 0.56–0.75 and the reconstructed-bin amplitude within 6% of the true-bin value, against the 0.52–0.76 and 5% it had kept from before it. The target-mass clause names which $\delta\Delta R$ it is measured against (8.7–40× the toy-input statistics the bound is computed with, 8.5–36× the grid-input $\delta\Delta R$ the reach column leads with). The $b_1$ transfer is given in full |



— the per-deuteron→per-nucleon  $\frac{1}{2}$ , the rank-2 0.92 and the  $\frac{1}{3}$  dilution, 0.15 in all — since 0.31 is the factor applied to a  $b_1$  already per nucleon. §5.3 names the Cloët-Bentz-Thomas curve the ratios are read off (their Eq. 23). §5.4 and Table 3 carry the  $5 \times 40.8$  caveat on the  ${}^7\text{Li}$   $\alpha$ : the 97% is an angle-only routing there, and the measured per-band insertion leaves 57%, station 2 only. Table 2's caption says the triton gun was fired at  $R = 1.286$  against the table's physical-mass 1.2897. The footer names Reports 1–4, and this appendix now runs newest first as Reports 2–4 do.

2026-08-28 The run-plan share stated where §6 quotes the reaches: the observables want different spin states, different far-forward optics — the coherent channel of §5.2 the tagging optics with its luminosity penalty — and, between the polarized EMC effect and the tensor sector, different isotopes,  ${}^6\text{Li}$  and  ${}^7\text{Li}$  being separate fills and separate runs, each reach being quoted at the whole year in its own, so Table 3's reaches are per observable and cannot be added into one year; how a year is actually divided is a programme decision priced in plans/07 WP2, under which every statistical error grows as one over the square root of the share and a luminosity quoted as a reach does not move. No number here moves.

2026-08-28 The target-mass ( $\gamma^2$ ) term that  $A_{||} \approx D \cdot g_1/F_1$  drops implemented behind `InclusiveKernel(target_mass=True)`, default off, and bounded rather than left implicit: §4.2 names the factor  $1 + \gamma^2$  the  $\approx$  hides ( $\gamma^2 \leq 0.010$  rate-weighted over the window of Table 3), and Table 3's polarized-EMC row the bias it leaves behind —  $\Delta R$  high by 0.12 / 0.44 / 0.71 / 1.06% at  $x = 0.09 / 0.28 / 0.45 / 0.71$ , 8.7–40× below the toy-input  $\delta\Delta R$  of the same weights (8.5–36× against the grid-input one) and 2.8–25× below the polarimetry scale ( `evgen/scripts/target_mass_bound.py` ). No published number moves.

2026-08-28 The theory curves of §§5.1 and 5.3 and of Table 3 replaced by the published figures themselves, digitized from the papers' vector paths: the polarized and unpolarized EMC ratios of [21] and [53] and the deuteron  $b_1$  of [45] and [46]. The polarized-EMC discrimination stays in the valence region but halves, to  $0.84\sigma$  per bin at  $10 \text{ fb}^{-1}/u$  and  $2.7\sigma$  at  $100 \text{ fb}^{-1}/u$  in the best bin of the window in which the two calculations are comparable at all, and the  $b_1$  scenarios separate into one that is measurable and one that is not. The  ${}^6\text{Li}$   $b_1$  transfer is the rank-2 tensor dilution  $\frac{1}{3} \times 0.92$  rather than the rank-1 vector 0.87. The Cosyn-Weiss closed-form tagged tensor asymmetry (§5.4) is added as a validated limit of the generator.

2026-08-28 §5.4 and Table 3 restated on the per-configuration far-forward optics, replacing the retired single proton-derived 73 / 164  $\mu\text{rad}$  pair on which the  ${}^6\text{Li}$   $\alpha$  tag was published as 1.9% / 1.5%. §5.2's coherent acceptances are labelled as the circular form of the formula they follow, with the anisotropic envelope and the measured pot edge given beside them. Later the same day the pot aperture and the over-rigid destination were measured in the current ePIC geometry (Report 3 §6.1, Report 4 §6, plans/09 B1): the pot edge is 2.50 / 1.51 / 0.53 mrad against 2.0 / 1.35 / 1.03, so §5.2's coherent tag reads  $9 \times 10^{-10} / 6 \times 10^{-27} / 7 \times 10^{-14}$  and the binding constraint is the silicon at  $5 \times 40.8$  and the beam at  $18 \times 137.5$ ; the  ${}^6\text{Li}$   $\alpha$  tag is 1.9 / 1.7 / 2.6% at the Yellow Report divergences and 28–35% under the tagging optics; and the  ${}^7\text{Li}$  triton, routed as lost while  $R = 1.29$  had no window, is measured onto the Roman-Pot silicon in 60 of 60 events with a calorimeter deposit behind it and is tagged in 78 / 92 / 94%, which opens the  $\alpha + t$  double tag.

2026-08-27 Rewritten as a paper: abstract, a single logical thread (atom → nucleus → ion; polarization → collider; formalism → observables → requirements), references numbered in citation order, and every repeated fact stated once. All numbers restated from the current scripts at the  $\gamma$ -matched beam energies ( ${}^6\text{Li}$  at 40.8 / 99.5 / 137.5 GeV/u, plans/10): the sweet spots, amplitudes and errors of the  $\Delta$  projection, the R-model shifts, the  $b_1$  and polarized-EMC reach, and the coherent-channel acceptance, which with the Yellow Report divergences does not survive at IP6 with any published optics. The pre-2026-08-27 coherent and tagging results computed at the superseded rigidity-scaled energies (the measured-aperture survival at 20.5 GeV/u, the  ${}^6\text{Li}$   $\alpha$  tag at 50 GeV/u) are withdrawn; the 13.5% of a 0.20 GeV envelope is retained only as the reference tag of the projections, no longer as an IP6 optics.

2026-08-26 Development run 8 folded in: the PYTHIA 8 hadronic final state, the Roman-Pot aperture measured in the ePIC geometry,  $R$  from the R1998 fit, physical nuclear masses in the far-forward routing.

2026-08-25 References verified against the local copies in refs/ (two citations corrected); the spin-state ratio and bunch-by-bunch alternation as the  $\Delta$  requirement; the two-state error formula.

2026-08-21 The why-lithium, why-polarization and fixed-target-versus-collider material (now §§2–3) rewritten against primary sources, and the requirements table added.

2026-08-17 First version: the DIS and spin-1 formalism from first principles, and the four flagship measurements ( $b_1$ ,  $\Delta$ , the polarized EMC effect, spectator-tagged DIS) with their reach.

Data availability: the code, figures and reproduction instructions are in the PolarizedLithiumSim repository ( `docs/reproduction_manual.md` ); companion documents are Reports 1–4 of this series. Projections use scenario structure functions with explicit bands.